ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CCS5411 CLOUD COMPUTING

FINAL EXAMINATION

DATE: Wednesday 28th May 2025

TIME: 14:00hrs – 17:00hrs

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **TWO (2)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Case Study: Cloud Computing Implementation at Zambezi Financial Services Ltd.

Zambezi Financial Services Ltd. (ZFSL) is a mid-sized Zambian financial institution based in Lusaka, offering a range of banking, investment, and insurance services to over 300,000 clients across the country. Established in 2005, the company has rapidly expanded due to its focus on digital transformation, financial inclusion, and customer-centric innovation.

As ZFSL scaled its operations, it faced significant challenges related to data storage, operational inefficiencies, and high IT infrastructure costs. Legacy on-premises servers could not handle the surge in transactional data, while system downtime impacted customer trust and compliance with financial service delivery standards. Additionally, ZFSL needed to align its digital infrastructure with national legal frameworks, particularly the **Data Protection Act of 2021** and the newly enacted **Cyber Security Act of 2025**.

The management resolved to implement a hybrid **cloud computing solution**, combining public and private cloud services, to enhance flexibility, scalability, and data security. The implementation would prioritize compliance with Zambia's data governance and cybersecurity laws while supporting the institution's long-term digital strategy.

ZFSL's cloud computing implementation aimed to achieve the following:

- Improve data availability and business continuity.
- Lower IT maintenance costs and enhance scalability.
- Ensure secure handling of sensitive financial and personal data.
- Align with legal requirements under Zambian law.

The implementation was scheduled in phases:

i. **Assessment and Planning**: Internal and external audits were conducted to assess current IT assets and regulatory risks.

ii. **Vendor Selection**: ZFSL selected Microsoft Azure as the primary public cloud provider and partnered with a local data center for private cloud hosting, ensuring data residency within Zambia.

iii. **Data Migration**: A phased migration began with non-sensitive data to test cloud integration, followed by customer and financial data under stringent encryption and access control protocols.

iv. **System Integration and Testing**: Cloud-hosted systems were integrated with ZFSL's core banking platform, with pilot testing across four branches.

v. **Training and Change Management**: Staff underwent cloud systems training, with ongoing support and cybersecurity awareness workshops.

Data Protection Act of 2021 – Compliance Measures

The **Data Protection Act, 2021** mandates that personal data must be processed lawfully, securely, and transparently. ZFSL ensured compliance by:

- Establishing a **Data Protection Officer (DPO)** to oversee compliance.
- Implementing **data minimization** and **purpose limitation** strategies.
- Encrypting customer data both in transit and at rest using AES-256 standards.
- Securing **explicit consent** from customers for data processing and cloud storage.
- Conducting **Data Protection Impact Assessments (DPIAs)** before migrating personal data.

Cyber Security Act of 2025 – Compliance Measures

The **Cyber Security Act, 2025** introduced new responsibilities around critical information infrastructure protection, cybercrime mitigation, and mandatory incident reporting. ZFSL responded by:

- Adopting the **National Cybersecurity Framework (NCF)** for financial services.
- Setting up a **Cybersecurity Operations Centre (CSOC)** to monitor threats 24/7.
- Implementing **multi-factor authentication (MFA)**, role-based access, and endpoint detection systems.
- Reporting all cyber incidents to the **Zambia Computer Incident Response Team (Z-CIRT)** within 24 hours as required by law.

Benefits Achieved

After six months of implementation, ZFSL reported:

- **99.95% system uptime**, enhancing customer trust.
- A **35% reduction in IT infrastructure costs**.
- Improved **regulatory audit outcomes**.

- Enhanced **data analytics capabilities**, enabling better customer insights.

Risks Identified

Despite the success, several risks were identified:

- **Vendor Lock-in**: Risk of overdependence on a single cloud provider.
- **Data Breach Potential**: External threats targeting cloud endpoints.
- **Staff Errors**: Possibility of human error during data handling.

Mitigation Measures

To counter these risks:

- ZFSL adopted a **multi-cloud strategy** to diversify service providers.
- Routine **penetration testing** and security audits were scheduled quarterly.
- Staff were subjected to **phishing simulations** and training in secure data practices.

Conclusion

ZFSL's cloud computing initiative marks a major milestone in its digital transformation journey. By aligning with Zambia's **Data Protection Act (2021)** and **Cyber Security Act (2025)**, the company has safeguarded its customers' trust while future-proofing its operations. The successful deployment has positioned ZFSL as a model for other financial institutions in Zambia.

QUESTION 1

Given the ***Cloud Computing Implementation at Zambezi Financial Services Ltd Case Study***, you are required to provide the ***Technology Considerations*** for the institution to achieve the most optimal cloud computing solution by providing answers to the following:

a) Discuss the ***cloud computing implementation types*** and make a recommendation of the model type that will be most optimal for Cloud Computing Implementation at Zambezi Financial Services Ltd giving reasons for your selection. **(15 Marks)**

b) Provide further specifications on the ***implementation cloud architecture*** be applied to support ZFSL's the ***selected model type in Question 1 a)*** effectively by providing answers to the following:

   i. What ***specific features of cloud platforms*** like Microsoft Azure support scalability and integration? **(3 marks)**

ii. How would you ***configure a virtual private network (VPN)*** to securely connect cloud services? **(4 marks)**

iii. In which scenarios would ZFSL ***prioritize private cloud over public cloud for data storage***? **(4 marks)**

c) ZFSL will also need to ***apply data encryption techniques*** to secure sensitive customer data in the cloud. In this regard, provide clarification on the following concerns:

i. What are the ***key differences*** between ***encryption at rest*** and ***encryption in transit***, and where should each be used? **(4 marks)**

ii. Why are ***cryptographic protocols*** (e.g., AES, TLS, RSA) needed for financial data security in ZFSL's context? **(4 marks)**

iii. How would you implement ***key management*** to ensure authorized access only? **(4 marks)**

d) ZFSL will need to ***pay particular attention to the contemporary issues*** affecting Cloud Computing in today's business needs environment. Advise ZFSL by giving the three (3) possible challenges and three (3) recommendations for the successful implementation and success of the project **(12 marks each)**

(Total: 50 marks)

SECTION B: Attempt any TWO (2) questions in this section

QUESTION 2

The Cloud Computing Solution design will need to be compliant and guarantee secure data storage for ZFSL under the ***Data Protection Act of 2021***. With aid of a diagram discuss the following:

a) The ***components must be included in the design*** to meet the principles of data minimization, purpose limitation, and accountability. **(12 marks)**

b) How would you ***integrate user consent mechanisms*** into the data storage lifecycle? **(8 marks)**

c) What ***cloud-native tools or third-party applications*** can be embedded to enforce access control and auditing? **(5 marks)**

(Total: 25 marks)

QUESTION 3

The solution will also need to be designed for a real-time threat detection and response system compliant with the Cyber Security Act of 2025. To guarantee this, you will be required to provide answers to the following:

a) What components are essential for building a ***Cybersecurity Operations Centre (CSOC)*** tailored to ZFSL's scale? **(7 marks)**

b) Which ***cloud-based security information*** and ***event management (SIEM) tools*** would you recommend and why? **(10 marks)**

c) How would you automate ***incident response workflows*** to ensure compliance with mandatory 24-hour breach reporting? **(8 marks)**

(Total: 25 marks)

QUESTION 4

For the ZFSL Cloud Computing Solution Implementation, apply ***software deployment best practices*** to ensure ***seamless integration between existing on-premise systems*** and ***new cloud-hosted applications***.

a) What ***tools (e.g., API gateways, middleware)*** can facilitate communication between legacy and cloud-native systems? **(12 marks)**

b) How would you ***perform continuous integration/continuous deployment*** (CI/CD) in a regulated financial environment? **(8 marks)**

c) How can ***containerization (e.g., Docker, Kubernetes)*** be applied to improve system portability and reliability? **(5 marks)**

(Total: 25 marks)

END OF EXAMINATION